

SE LAB 3 JUSTIFICATION

Name: Janhavi Ramakrishnan

SRN: PES1UG24CS198

Selected Architecture: Layered Architecture

For the Online Doctor Consultation and Booking System, Layered Architecture was selected. The system is divided into three main layers: Presentation Layer, Business Layer, and Data Layer. The Presentation Layer contains the Patient/Doctor UI, the Business Layer contains the core system components, and the Data Layer contains the Database.

Reason 1 – Separation of Responsibilities

Layered Architecture provides a clear separation of responsibilities between different parts of the system. The Presentation Layer handles interaction with patients and doctors, while the Business layer handles functions such as doctor and availability management, appointment booking, online consultations, and prescription management. The Data Layer is responsible for storing and retrieving system data. This separation makes the system easier to understand, develop, test, and maintain.

Reason 2 – Easier Maintenance and Modification

The layered structure allows changes to one part of the system to be made with less impact on the other layers. For example, changes to the user interface can be made without directly modifying the database. Similarly, business functionality such as appointment booking or prescription management can be modified while keeping the presentation layer separate. This makes the system easier to maintain and extend as new features are added.

Security Advantage

Layered Architecture improves security by preventing the user interface from directly accessing the database. Requests from the Patient/Doctor UI pass through the business components before accessing stored data. This allows access to sensitive information such as patient, consultation, and prescription data to be controlled through the business layer. The separation of the Data Layer also reduces direct exposure of the database to users.

Performance Benefit

The architecture separates frequently used operations into individual business components, such as doctor availability, appointment booking, and online consultation. This allows each operation to process only the data and functionality required for that request. Database access can also be handled through the business layer, avoiding unnecessary direct communication between the user interface and database.